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Quantum-Inspired Secure Watermark Encryption Framework for Video Authentication

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ABSTRACT:

The rapid growth of digital video content across surveillance systems, social media platforms, and multimedia communication channels has raised significant concerns regarding video authenticity and integrity. With the availability of advanced video editing tools and artificial intelligence technologies, it has become increasingly easy to manipulate video content. Such modifications can compromise the reliability of digital media used in legal evidence, journalism, and security applications. Digital watermarking is widely used as a technique to verify multimedia authenticity by embedding hidden information within the media content. However, many existing watermarking systems embed plaintext watermark information without strong encryption mechanisms. This exposes watermark data to various attacks such as watermark extraction, replay attacks, and statistical analysis. Therefore, securing watermark information before embedding is essential to enhance the reliability of watermark-based authentication systems.

This project proposes a quantum-inspired secure watermark framework designed to protect watermark information before it is embedded into video frames. The framework employs chaotic key generation techniques inspired by quantum randomness. A dynamic encryption key is derived from a combination of a user secret, video content hash, and frame index, ensuring that each video frame receives a unique encryption key. The watermark is converted into binary bits and encrypted through a series of operations including key-dependent permutation and XOR-based diffusion, introducing strong confusion and diffusion properties.

To evaluate the security strength of the proposed encryption scheme, several experimental modules were implemented, including key sensitivity analysis, entropy analysis, and correlation analysis. The experimental results demonstrate that the encrypted watermark exhibits high randomness, strong key sensitivity, and minimal correlation with the original watermark. The proposed framework can serve as a secure watermark protection layer that can be integrated with existing video watermarking systems for reliable real-time video authentication.

Keywords: Video Authentication, Watermark Encryption, Chaotic Encryption, Quantum-Inspired Security, Multimedia Security

[1.] INTRODUCTION:

Digital video has become an essential medium for communication, surveillance, and information exchange in modern society. Video data is extensively used in applications such as security monitoring, forensic investigations, social media content distribution, and online broadcasting. However, the widespread availability of powerful video editing software and artificial intelligence technologies has significantly increased the risk of digital video manipulation. Even minor modifications to video frames can compromise the credibility of multimedia content and lead to misinformation. Digital watermarking is one of the most widely used methods for protecting multimedia authenticity. In watermarking, hidden information is embedded into digital media in such a way that it can later be extracted to verify content ownership or integrity. While watermarking techniques can effectively detect modifications in multimedia content, many existing systems embed watermark information directly without encryption. As a result, attackers may analyze watermark patterns, extract watermark information, or reuse watermarks across different videos. To improve the security of watermark-based authentication systems, watermark information must be protected before it is embedded into the media content. Encryption techniques can prevent unauthorized extraction and interpretation of watermark data. However, traditional encryption approaches may not provide sufficient randomness or dynamic behaviour when applied to watermark data in video streams.

In this project, a quantum-inspired secure watermark framework is proposed to address these limitations. The framework uses chaotic encryption techniques inspired by quantum randomness to generate secure watermark representations. Dynamic encryption keys are generated using a combination of user secret, video hash, and frame index, ensuring that watermark encryption varies across frames and videos. The watermark bits undergo permutation and diffusion operations to achieve strong confusion and diffusion properties. The proposed framework focuses on enhancing the security of watermark data and can be integrated with any watermark embedding system to improve video authentication reliability.

[2.] PROBLEM STATEMENT:

With the increasing use of digital video across various domains such as surveillance, media broadcasting, and forensic investigation, ensuring the authenticity and integrity

of video content has become a critical challenge. Modern video editing tools and artificial intelligence technologies enable attackers to easily manipulate video frames, making it difficult to determine whether a video has been altered.

Digital watermarking techniques are commonly used to detect modifications in multimedia content. However, many watermarking systems embed watermark information directly without sufficient encryption mechanisms. This makes the watermark vulnerable to attacks such as watermark extraction, replay attacks, and statistical analysis. Attackers may analyze watermark patterns, extract watermark bits, or reuse watermark information across different videos.

Another limitation of existing watermark security mechanisms is the use of static encryption keys. Static keys can make watermark systems vulnerable to key-guessing attacks and replay attacks, where watermark information from one frame or video can be reused in another.

Therefore, there is a need for a secure watermark protection framework that can dynamically encrypt watermark information before it is embedded into video frames. The encryption mechanism should generate highly unpredictable watermark representations, exhibit strong key sensitivity, and resist statistical attacks. Such a framework would significantly improve the security of watermark-based video authentication systems.

[3.] OBJECTIVES:

The primary objectives of this project are:

1. To design a secure watermark encryption framework for protecting watermark information used in video authentication systems.
2. To develop a quantum-inspired chaotic encryption mechanism capable of generating highly unpredictable watermark representations.
3. To generate dynamic encryption keys using a combination of user secret, video content hash, and frame index.
4. To implement permutation and diffusion operations to enhance watermark encryption security.
5. To evaluate the security properties of the proposed encryption scheme using

experimental analyses such as key sensitivity, entropy analysis, and correlation analysis.

6. To demonstrate that the proposed framework can serve as a secure watermark protection layer for real-time video authentication systems.

[4.] SCOPE OF THE PROJECT:

The scope of this project focuses on the design and implementation of a secure watermark encryption framework for video authentication systems. The proposed framework aims to enhance the security of watermark information before it is embedded into video frames. The project involves the development of a quantum-inspired chaotic encryption scheme that converts watermark images into encrypted watermark representations. The encryption mechanism introduces strong randomness, key sensitivity, and resistance to statistical analysis. The system generates dynamic encryption keys based on a combination of user secret, video content hash, and frame index, ensuring that watermark encryption varies across frames and videos. The project also includes the implementation of several experimental modules to evaluate the security strength of the encryption scheme. These modules analyze key sensitivity, entropy, and correlation properties of the encrypted watermark. The proposed framework is designed as a security layer that can be integrated with existing watermark embedding systems. The project does not focus on the design of the watermark embedding algorithm itself but rather on enhancing the security of watermark information before embedding.

[5.] PROPOSED SYSTEM:

The proposed system introduces a secure watermark encryption framework that protects watermark information before it is embedded into video frames. The framework is designed to enhance the security and reliability of watermark-based video authentication systems.

The system begins by generating a watermark logo that represents authentication information such as ownership details or verification data. The watermark image is converted into a binary bitstream to enable cryptographic processing.

Next, a dynamic encryption key is generated using a combination of three parameters:

a user secret key, a video content hash, and the frame index. The user secret serves as the primary private key known only to the system owner. The video hash binds the watermark encryption to a specific video, preventing watermark reuse across different videos. The frame index ensures that each frame receives a unique encryption key, preventing replay attacks between frames.

Once the encryption key is generated, the watermark bits undergo a permutation process based on the generated key. This permutation rearranges the order of the watermark bits, disrupting the original structure of the watermark data. Following permutation, XOR-based diffusion is applied between the permuted watermark bits and the encryption key. This diffusion step ensures that small changes in the input watermark or key produce significant changes in the encrypted output.

The resulting encrypted watermark appears statistically random and bears minimal similarity to the original watermark. This encrypted watermark can then be embedded into video frames using any watermark embedding system.

During verification, the encrypted watermark extracted from the video is decrypted using the same chaotic key generation process. The decrypted watermark is then compared with the original watermark to determine the authenticity of the video content.

[6.] LITERATURE SURVEY:

S. No	Name of Paper	Type of Paper	Authors / Organization	Core Contribution	Research Gap Addressed by Present Invention
1	Deep learning-based watermarking techniques: challenges and future trends	Journal (Springer)	S. Ben Jabra, M. Ben Farah	Comprehensive review of CNN-based image & video watermarking	Existing works focus on watermark embedding using deep learning but lack dedicated encryption frameworks to secure the watermark information before embedding.

2	Digital Watermarking Using Artificial Intelligence: Techniques and Trends	Journal (Wiley)	A. Dandooh, A.S. El-Fishawy	Survey of AI, CNN, autoencoder watermarking methods	Current AI-based watermarking research emphasizes model architectures while ignoring secure watermark encryption and dynamic key-based authentication mechanisms.
3	Modeling Automated Image Watermarking Using Meta-Heuristic Deep CNN	Journal (Springer)	L.R. Battarusetty et al.	CNN optimized via chaotic/metaheuristic algorithms	Chaotic optimization is applied to improve CNN performance, but watermark data itself is not encrypted, leaving the watermark vulnerable to unauthorized extraction.
4	Fusion Model of Quantum Wavelet Transform and Neural Network	Conference / Journal	I. Alhakam et al.	Hybrid quantum wavelet + NN for video security	Quantum-inspired processing is used for feature extraction but does not provide a secure encryption framework for watermark protection and authentication.
5	Robust Image Watermarking using DWT, DCT and CNN-Based Attack Evaluation	Journal (IAJIT)	M. Alia et al.	CNN used to evaluate robustness of watermark	The approach focuses on robustness against attacks but does not secure the watermark itself through encryption prior to embedding.
6	Survey on Deep Learning Applications in Digital Image Security	Journal (SPIE)	Z. Bao, R. Xue	Covers CNN watermarking, attacks & defenses	Most approaches concentrate on image security techniques rather than developing a dedicated security framework for encrypted watermark-based video authentication.

7	Recent Trends in Image Watermarking Techniques for Copyright Protection	Journal (Springer)	A. Ray, S. Roy	Survey including quantum watermarking	Although various watermarking approaches are discussed, secure encryption mechanisms integrated with watermarking systems remain limited.
8	Chaotic Image Security Techniques and Developments: A Review	Journal (MDPI)	H. Zhang et al.	Review of chaos & quantum watermarking	Chaotic encryption techniques are studied separately, but their integration with watermark-based video authentication frameworks is insufficient.
9	A Fragile Image Watermarking Scheme Using Quantum Chaotic Maps	Journal (MDPI)	A.M. Ramos et al.	Quantum-chaos-based watermarking	The fragile watermarking scheme is suitable for tamper detection but lacks encryption-based protection and scalability for real-time video environments.
10	Neural Network-Based Prediction of Möbius Transform Parameters	Preprint (SSRN)	A. Alrammahi, H. Sajedi	CNN predicts transform parameters	The work focuses on transform parameter prediction but does not address secure watermark encryption or authentication for multimedia data.
11	The Influence of Neural Network Image Compression on Watermarks	IEEE Conference	M. Yakubenko, M. Gashnikov	Impact of NN compression on watermark survival	The study evaluates compression effects on watermark robustness but lacks mechanisms for encrypting watermark information to prevent unauthorized decoding.

12	Distinguishing Reality from AI: Detecting Synthetic Content	Journal (MDPI)	D. Ghiurău et al.	CNN-based detection of watermarked media	Detection-based methods focus on identifying manipulated media rather than securing watermark information through encryption-based authentication frameworks.
13	Enhancing Visual Perception in Real-Time Using Deep RL	Journal	S.T. Chopparapu et al.	Reinforcement learning for real-time visuals	Although real-time processing is addressed, watermark security and authentication mechanisms are not integrated into the system.
14	A Novel Framework for Digital Image Watermarking Based on NN	Journal	J. He	Lightweight NN watermarking model	The proposed framework focuses on efficient watermark embedding but does not include encryption modules to protect watermark data from unauthorized access.
15	Hypercomplex Techniques in Signal & Image Processing	IEEE Journal	A. Alcayde, J. Ventura	Quaternion CNNs for signal processing	Advanced signal processing techniques are proposed, but their application in secure watermark encryption frameworks is not explored.
16	Deep Learning-Driven Multi-layered Steganographic Approach	Journal (Nature Sci Rep)	Y. Sanjalawe et al.	CNN-based multi-layer embedding	The approach emphasizes data concealment rather than secure watermark encryption and authentication for digital media verification.

17	Watermarking for Secure IoT Video Transmission	Conference	Various IEEE Authors	Focus on IoT video watermarking	IoT-based watermarking techniques address transmission security but lack encryption-based watermark frameworks for verifying video authenticity.
18	Chaotic Watermarking for Tamper Detection	Book Chapter	H. Kaur	CNN + chaos for tamper detection	Chaotic watermarking methods detect tampering but do not provide a structured encryption framework for secure watermark generation and verification.
19	Literature Survey of Image Forgery Detection Using ML	Thesis / Survey	S. Bipasha	CNN detection of watermark distortions	Most works focus on detecting forged images rather than implementing secure watermark encryption mechanisms for proactive authenticity verification.
20	Ensuring Digital Image Authenticity Using Blockchain & Watermarking	IEEE Conference	V.P. Bagdi et al.	Blockchain + watermarking	Blockchain-based authenticity verification is explored, but watermark encryption and lightweight security frameworks suitable for real-time video are not addressed.

[6.1] FINDINGS IN LITERATURE SURVEY:

The literature survey on digital watermarking, multimedia security, and deep learning-based watermarking techniques reveals several important observations regarding the current state of research in the field of digital media authentication.

Firstly, a large number of studies focus on improving watermark embedding and extraction techniques using deep learning models such as Convolutional Neural Networks (CNNs), autoencoders, and other machine learning approaches. These techniques primarily aim to

enhance watermark robustness against various signal processing attacks including compression, noise addition, scaling, and filtering. While these approaches significantly improve watermark survivability, most of them do not focus on securing the watermark data itself before embedding, which creates a potential vulnerability if an attacker manages to extract or analyze the watermark information.

Secondly, several research works explore chaotic systems and quantum-inspired techniques for image encryption and multimedia security. These techniques demonstrate strong cryptographic properties such as high sensitivity to initial conditions, strong randomness, and diffusion characteristics. However, in many cases these encryption techniques are studied independently of watermarking frameworks, meaning they are not integrated with watermark generation and authentication pipelines for multimedia verification.

Another important observation is that existing watermarking frameworks often assume that the watermark is inherently secure, without considering the possibility that the watermark itself can be intercepted, copied, or reverse-engineered. In real-world applications such as digital content protection, video authentication, and copyright verification, it is critical that the watermark information remains secure even if the embedding algorithm is known. Therefore, protecting the watermark through encryption becomes an essential requirement.

Furthermore, many existing watermarking systems focus on robust watermarking, where the goal is to preserve watermark information even after heavy distortions. However, these systems rarely incorporate authentication-oriented watermark security mechanisms that allow reliable verification of digital media integrity. This limitation highlights the need for frameworks that combine watermarking with secure cryptographic protection.

Additionally, several studies discuss watermarking in the context of IoT systems, multimedia streaming, and digital content distribution, but they often overlook the importance of lightweight encryption frameworks suitable for real-time video environments. Real-time applications require solutions that provide strong security while maintaining computational efficiency.

Based on these observations, it is evident that there is a significant research gap in the integration of watermark encryption frameworks with multimedia authentication systems. Most existing approaches either focus on watermark embedding techniques or on standalone encryption methods, but very few works combine both aspects into a unified

framework.

To address these limitations, the present work proposes a Quantum-Inspired Secure Watermark Framework for Video Authentication, which introduces a structured encryption mechanism for watermark protection. The framework incorporates chaotic key generation, permutation-based bit mixing, and XOR-based diffusion operations to securely transform watermark data before embedding. This approach enhances the security of watermark information and strengthens the overall reliability of video authentication systems.

By integrating encryption, watermark processing, and security evaluation modules within a unified architecture, the proposed framework aims to improve watermark confidentiality, resistance against unauthorized extraction, and robustness of video authenticity verification mechanisms.

[7.] METHEDOLOGY:

The methodology of the proposed system consists of several stages including watermark preparation, key generation, watermark encryption, and security evaluation.

Initially, a watermark image is generated to represent authentication information. The watermark image is converted into a binary bitstream so that cryptographic operations can be applied to the watermark data.

Next, a dynamic encryption key is generated using a quantum-inspired chaotic process. The key is derived from three parameters: a user secret, a video hash, and a frame index. These parameters are combined and processed using a hashing function to generate an initial seed value. The seed value is then used to generate a chaotic sequence using the logistic map equation. This chaotic sequence forms the encryption key used in subsequent operations.

After key generation, the watermark bits undergo a permutation process. The permutation operation rearranges the watermark bits according to the generated encryption key, effectively disrupting the spatial structure of the watermark. This step introduces confusion in the encryption process, making it difficult for attackers to infer the original watermark structure.

Following permutation, a diffusion process is applied using XOR operations between the permuted watermark bits and the encryption key. This diffusion ensures that small changes in the input watermark or key propagate across the encrypted watermark, producing significant changes in the ciphertext.

Finally, the security strength of the encryption scheme is evaluated using several experimental modules including key sensitivity analysis, entropy analysis, and correlation analysis.

[8.] SOFTWARE REQUIREMENTS:

Functional Requirements

1. The system must generate a watermark image for authentication.
2. The system must convert the watermark image into binary representation.
3. The system must generate dynamic encryption keys based on user secret, video hash, and frame index.
4. The system must perform permutation and diffusion operations to encrypt watermark bits.
5. The system must support watermark decryption for verification.
6. The system must perform security analysis experiments to evaluate encryption strength.

Non-Functional Requirements

1. The encryption process should produce highly random watermark representations.
2. The system should exhibit strong key sensitivity to prevent key-guessing attacks.
3. The encrypted watermark should have minimal correlation with the original watermark.
4. The encryption process should be computationally efficient to support real-time applications.

[9.] SYSTEM ARCHITECTURE:

The proposed system architecture is designed to provide a secure watermark protection framework for video authentication. The architecture focuses on encrypting watermark information before it is embedded into video frames, thereby protecting watermark data from unauthorized extraction, replay attacks, and statistical analysis.

The overall architecture consists of four main modules:

1. Watermark Generation Module
2. Chaotic Key Generation Module
3. Quantum-Inspired Watermark Encryption Module
4. Watermark Verification Module

Each module performs a specific function in the secure watermark framework, and together they ensure the confidentiality and integrity of watermark information used for video authentication.

[9.1] WATERMARK GENERATION MODULE

The watermark generation module is responsible for creating the watermark information that will be used for authentication. The watermark typically consists of a logo, symbol, or binary pattern representing ownership or verification data. The watermark image is first converted into a binary representation, where pixel values are transformed into a sequence of binary bits. This conversion is necessary because encryption algorithms operate on binary data rather than image pixels. The resulting binary watermark sequence serves as the input for the encryption module. At this stage, the watermark still retains its original structure and patterns, which makes it vulnerable to statistical analysis if embedded directly. Therefore, encryption is required to secure the watermark before embedding.

[9.2] CHAOTIC KEY GENERATION MODULE

The chaotic key generation module is responsible for generating a dynamic encryption key used for watermark encryption. Instead of using static keys, the proposed framework generates encryption keys based on multiple parameters to enhance security.

The key generation process combines the following three inputs:

• User Secret Key – A private key known only to the system owner.
• Video Content Hash – A hash value generated from the input video, ensuring that the encryption is tied to a specific video.
• Frame Index – The position of the frame within the video sequence.

These three parameters are combined to generate a seed value using a cryptographic hash function. The seed value is then used to generate a chaotic sequence using a logistic map equation, which is a well-known chaotic function used in cryptographic applications. The logistic map generates a sequence of values that exhibit high sensitivity to initial conditions, meaning that even a small change in the seed value produces a completely different chaotic sequence. This property ensures strong key sensitivity and unpredictability in the encryption process. The chaotic sequence produced by the logistic map is converted into a binary key sequence that is used during the watermark encryption stage.

[9.3] QUANTUM-INSPIRED WATERMARK ENCRYPTION MODULE

The encryption module is the core component of the proposed framework. Its purpose is to transform the original watermark bits into an encrypted watermark representation that appears statistically random and cannot be interpreted without the correct key.

The encryption process consists of two main operations:

9.3.1 Permutation Process

The permutation step rearranges the order of the watermark bits using the generated chaotic key sequence. The permutation pattern is determined by the chaotic key, meaning that the bit positions are shuffled in a key-dependent manner. This permutation step introduces confusion into the encryption process. By rearranging the watermark bits, the spatial structure of the original watermark image is destroyed. As a result, attackers cannot infer the original watermark pattern from the encrypted watermark. Permutation is particularly important because it prevents attackers from identifying recognizable watermark shapes or structures in the encrypted data.

9.3.2 XOR Diffusion Process

After the permutation process, the watermark bits undergo a diffusion operation. In this step, the permuted watermark bits are combined with the chaotic key sequence using an XOR operation. The XOR diffusion step ensures that small changes in the watermark bits or encryption key propagate across the entire encrypted watermark sequence. This property is known as diffusion, and it significantly improves the security of the encryption process. The combination of permutation and diffusion ensures that the encrypted watermark exhibits strong randomness and minimal statistical correlation with the original watermark. The output of this module is an encrypted watermark sequence, which appears as random binary data and contains no visible information about the original watermark.

[9.4] ENCRYPTED WATERMARK OUTPUT

The encrypted watermark sequence generated by the encryption module can now be embedded into video frames using a watermark embedding system. Since the watermark has already been encrypted, even if an attacker successfully extracts the watermark from the video frames, they will only obtain the encrypted watermark representation. Without the correct decryption key and chaotic key generation process, it will be extremely difficult to recover the original watermark. Thus, the encryption framework provides an additional layer of security that protects watermark information against unauthorized extraction and analysis.

[9.5] WATERMARK DECRYPTION AND VERIFICATION MODULE

The verification module is responsible for recovering and validating the watermark information during the authentication process. When a video is received for verification, the watermark is first extracted from the video frames using the watermark extraction mechanism. The extracted watermark at this stage is still in encrypted form. The system then regenerates the chaotic encryption key using the same parameters used during encryption, namely the user secret, video hash, and frame index.

Using this regenerated key, the encrypted watermark undergoes two reverse operations:

• Reverse XOR Diffusion – The XOR operation is reversed to recover the permuted watermark bits.
• Reverse Permutation – The bit positions are restored to their original order using the

inverse permutation sequence.

Once these steps are completed, the system reconstructs the original watermark image from the decrypted watermark bits. The recovered watermark is then compared with the original watermark to verify the authenticity of the video. If the watermark matches the original watermark, the video is considered authentic. Otherwise, the video may have been tampered with or manipulated.

[9.6] SECURITY EVALUATION MODULES

To evaluate the effectiveness of the proposed encryption framework, several experimental modules were implemented:

1. Key Sensitivity Analysis - This experiment measures how small changes in the secret key affect the encrypted watermark. The results show that a minor change in the key results in significant changes in the encrypted watermark, demonstrating strong chaotic behaviour.
2. Entropy Analysis - Entropy measures the randomness of the encrypted watermark. The results indicate that the encrypted watermark has entropy values close to the theoretical maximum, confirming high randomness.
3. Correlation Analysis - This experiment measures the similarity between the original watermark and the encrypted watermark. The correlation values obtained are close to zero, indicating that the encrypted watermark has minimal similarity with the original watermark.

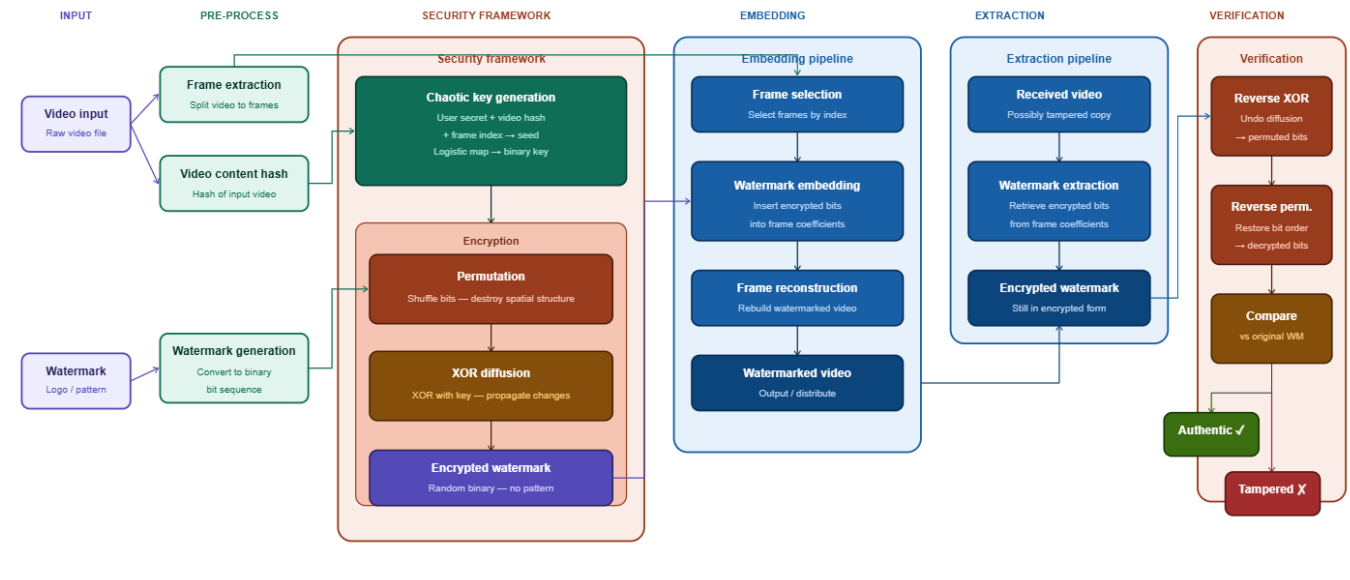

Fig. 9.1 System Architecture of the Proposed Video Authentication and Watermarking Framework

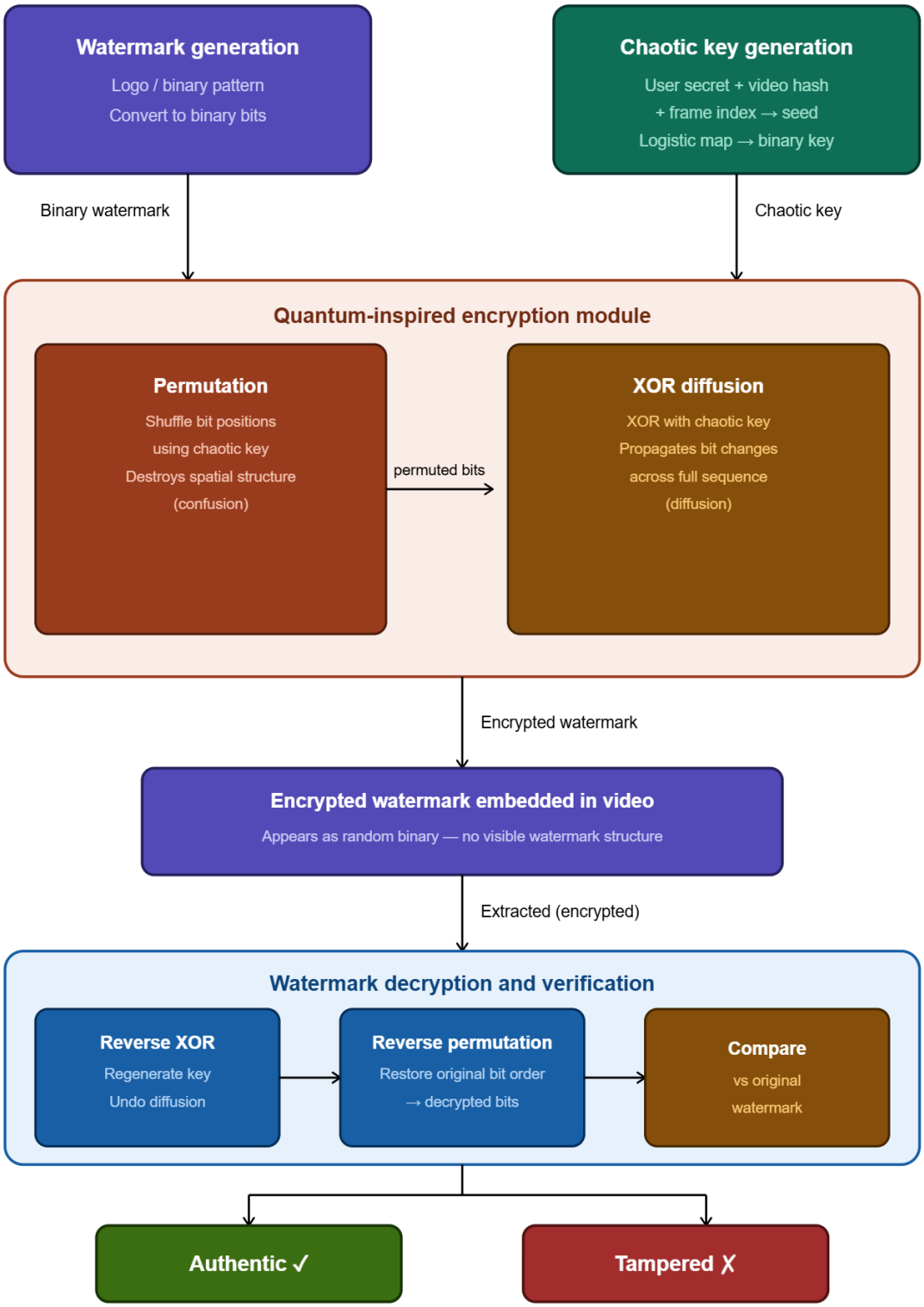

Fig. 9.2 Quantum-Inspired Secure Watermark Encryption Framework

[10]. UML DIAGRAMS:

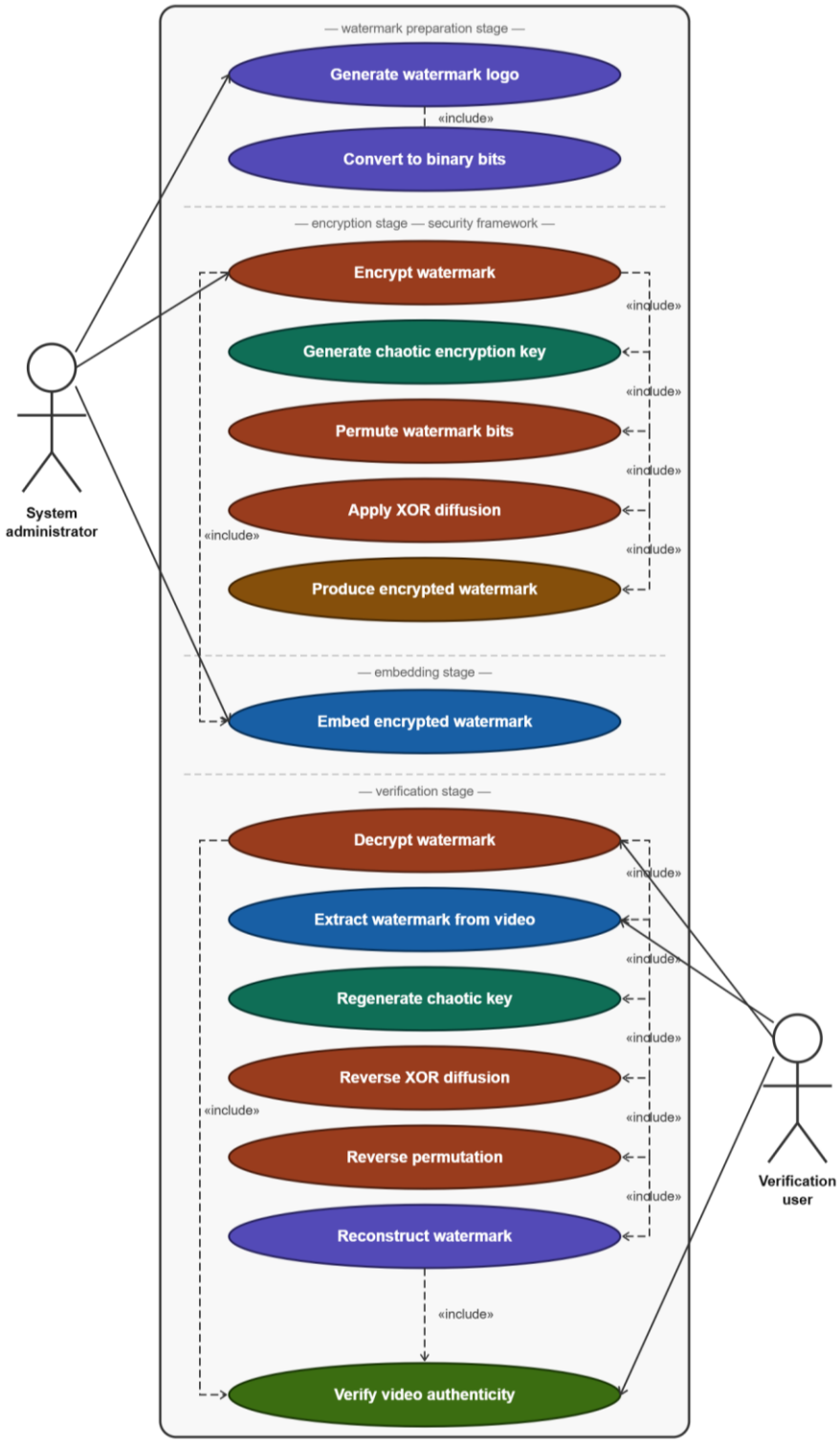

Fig. 10.1 UML Use Case Diagram of the Quantum-Inspired Secure Watermark Framework for Video Authentication

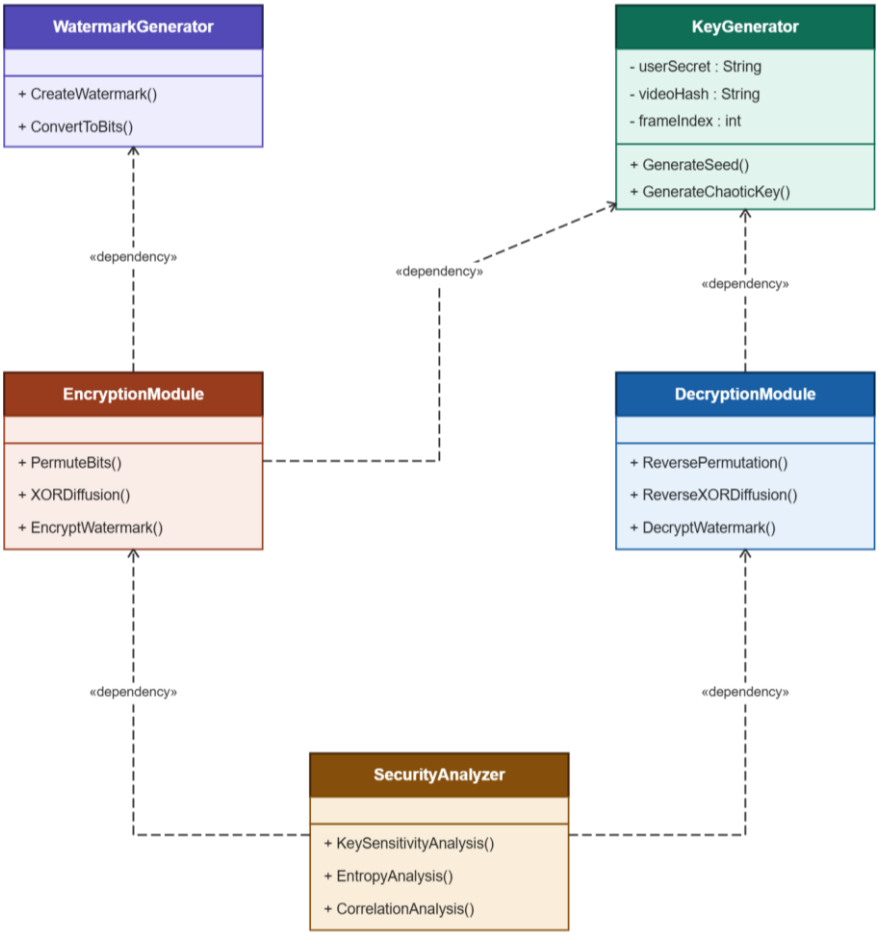

Fig. 10.2 UML Class Diagram of the Proposed Secure Video Watermarking System

[11] IMPLEMENTATION:

The proposed Quantum-Inspired Secure Watermark Framework for Video Authentication was implemented using a modular architecture in Python. The implementation focuses on securing watermark information before embedding by applying a quantum-inspired encryption process. The system consists of several functional modules including watermark generation, chaotic key generation, watermark encryption, watermark decryption, and security evaluation. The entire framework was developed in a structured manner so that each component operates independently while interacting with other modules through well-defined interfaces. This modular design improves system flexibility and allows the encryption framework to be integrated with various watermark embedding techniques.

[11.1] DEVELOPMENT ENVIRONMENT

The implementation of the proposed framework was carried out using the following development tools and technologies.

Component	Tool / Technology
Programming Language	Python
Development Environment	Visual Studio Code
Image Processing	OpenCV
Numerical Computation	NumPy
Visualization	Matplotlib
Operating System	Windows

[11.2] WATERMARK GENERATION MODULE

The watermark generation module is responsible for creating the watermark logo and converting it into a binary format suitable for encryption. In this module, a watermark image is generated and stored as a grayscale image. The grayscale image is then processed to extract pixel values, which are converted into binary bit sequences. These binary bits represent the watermark information that will later be encrypted before embedding into the video frames. The steps involved in this module include:

1. Creation of a watermark logo image.
2. Conversion of the watermark image into grayscale format.
3. Extraction of pixel intensity values.
4. Conversion of pixel values into binary bit representation.

The generated bit sequence serves as the input for the encryption module.

[11.3] CHAOTIC KEY GENERATION MODULE

To improve the security of the watermark, the proposed framework uses a chaotic key generation mechanism. The key generator produces dynamic encryption keys based on contextual parameters related to the video and the user. The key generation process uses the following inputs:

- User secret key
- Video hash value
- Frame index

These parameters are combined to generate a seed value, which is then used to generate a

chaotic key sequence. The chaotic key sequence introduces randomness into the encryption process and ensures that the encryption pattern varies for different video frames. This dynamic key generation approach provides several advantages:

- Increased security against brute-force attacks
- High sensitivity to input parameters
- Unique encryption keys for different video frames

The same key generation process is used during both encryption and decryption to maintain consistency.

[11.4] WATERMARK ENCRYPTION MODULE

The watermark encryption module is the core component of the proposed security framework. This module applies a quantum-inspired encryption process to the watermark bits before embedding them into the video. The encryption process consists of two main stages:

1. Permutation-Based Bit Mixing

In the first stage, the watermark bits are rearranged using a key-dependent permutation sequence generated from the chaotic key. This step changes the order of the watermark bits, making it difficult for an attacker to determine the original watermark structure. This process mimics the concept of quantum superposition, where information does not have a fixed state until it is observed.

2. XOR-Based Diffusion

In the second stage, the permuted watermark bits undergo diffusion using a bitwise XOR operation with the chaotic key sequence. The XOR diffusion spreads the influence of the key across the entire watermark. As a result, even a small change in the encryption key leads to a completely different encrypted watermark.

After the permutation and diffusion stages, the system produces the encrypted watermark, which can then be embedded into video frames.

[11.5] WATERMARK DECRYPTION MODULE

The watermark decryption module performs the inverse operations of the encryption process in order to recover the original watermark during verification. The decryption process includes the following steps:

1. Regeneration of the chaotic key using the same input parameters.
2. Application of reverse XOR diffusion.
3. Reversal of the permutation operation.
4. Reconstruction of the watermark from the recovered bit sequence.

If the correct key parameters are used, the original watermark is successfully reconstructed. If incorrect parameters are used, the recovered watermark becomes distorted, enabling detection of unauthorized access or tampering.

[11.6] SECURITY EVALUATION MODULE

To evaluate the security strength of the proposed encryption framework, several security analysis experiments were conducted. These experiments analyze the robustness and randomness of the encrypted watermark. The following tests were implemented.

1. Key Sensitivity Analysis: Key sensitivity analysis evaluates how sensitive the encryption algorithm is to small changes in the encryption key. A secure encryption algorithm should produce significantly different encrypted outputs even if the key changes slightly.

2. Entropy Analysis: Entropy measures the randomness of the encrypted watermark. Higher entropy values indicate better randomness and stronger resistance against statistical attacks.

3. Correlation Analysis: Correlation analysis measures the similarity between the original watermark and the encrypted watermark. A low correlation value indicates that the encrypted watermark does not reveal information about the original watermark.

These security evaluation metrics confirm that the proposed encryption framework provides strong protection against unauthorized watermark extraction.

[11.7] EXPERIMENTAL EXECUTION

The experimental evaluation of the proposed framework was carried out using watermark images generated within the system. The watermark image was first converted into binary bits and then encrypted using the proposed quantum-inspired encryption algorithm. The encrypted

watermark was then subjected to security analysis tests including key sensitivity analysis, entropy measurement, and correlation analysis. The results obtained from these experiments demonstrate that the proposed framework achieves strong watermark protection and high resistance to statistical attacks. The modular implementation also ensures that the encryption framework can be integrated with various watermark embedding systems for secure video authentication.

[12] SUMMARY:

This project presents a quantum-inspired secure watermark framework designed to enhance the security of watermark-based video authentication systems. The proposed approach focuses on encrypting watermark information before it is embedded into video frames, thereby preventing unauthorized extraction and interpretation of watermark data. The framework employs chaotic key generation techniques combined with permutation and diffusion operations to produce encrypted watermark representations with strong randomness and key sensitivity. Dynamic encryption keys derived from user secret, video hash, and frame index ensure that watermark encryption varies across frames and videos. Experimental evaluation using key sensitivity analysis, entropy analysis, and correlation analysis demonstrates that the proposed encryption scheme provides strong security properties and resistance to statistical attacks. The proposed framework can be integrated with existing watermark embedding systems to provide a secure watermark protection layer for real-time video authentication applications.

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